

Chapter 22

Noise Pollution

The ability to make and detect sound provides humans with the ability to communicate with each other as well as to receive useful information from the environment. Sound can provide warning, as in the fire alarm, information, as from a whistling tea kettle, and enjoyment, as from music. In addition to such useful and pleasurable sounds there is noise, often defined as unwanted or extraneous sound. We generally classify the unwanted sound made by such products of our civilization as trucks, airplanes, industrial machinery, air conditioners, and similar sound producers as noise.

Damage to the ear is a measurable problem associated with noise; annoyance, while not necessarily easily measured, may be of equal concern. In one Danish investigation of annoyance factors in the workplace, 38% of workers interviewed identified noise as the most annoying factor in their environment.

Urban noise is not, surprisingly, a modern phenomenon. Legend has it, for example, that Julius Caesar forbade the driving of chariots on Rome's cobblestone streets during nighttime so he could sleep (Still 1970). Before the ubiquitous use of automobiles, the 1890 noise pollution levels in London were described by an anonymous contributor to the *Scientific American* (Taylor 1970) as:

The noise surged like a mighty heart-beat in the central districts of London's life. It was a thing beyond all imaginings. The streets of workaday London were uniformly paved in "granite" sets . . . and the hammering of a multitude of iron-shod hairy heels, the deafening side-drum tattoo of tyred wheels jarring from the apex of one set (of cobblestones) to the next, like sticks dragging along a fence; the creaking and groaning and chirping and rattling of vehicles, light and heavy, thus maltreated; the jangling of chain harness, augmented by the shrieking and bellowing called for from those of God's creatures who desired to impart information or proffer a request vocally — raised a din that is beyond conception. It was not any such paltry thing as noise.

Noise may adversely affect humans both physiologically as well as psychologically. It is an insidious pollutant; damage is usually long range and permanent.

THE CONCEPT OF SOUND

The average citizen has little, if any, concept of what sound is or what it can do. This is exemplified by the tale of a well-meaning industrial plant manager who decided

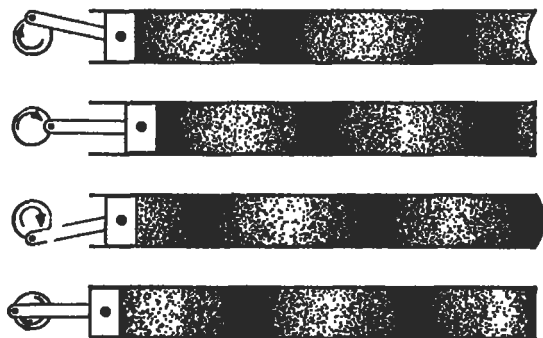


Figure 22-1. A piston creates pressure waves that are transmitted through air.

to decrease the noise level in his factory by placing microphones in the plant and channeling the noise through loudspeakers to the outside (Lipscomb 1975).

Sound is transfer of energy without transfer of mass. For example, a rock thrown at you would certainly get your attention, but this would require the transfer of the rock's mass. Alternatively, your attention may be gained by poking you with a stick, in which case the stick is not lost, but energy is transferred from the poker to the target. In the same way, sound travels through a medium such as air without a transfer of mass. Just as the stick must move back and forth, so must air molecules oscillate in waves to transfer energy.

The small displacement of air molecules that creates pressure waves in the atmosphere is illustrated in Fig. 22-1. As the piston is forced to the right in the tube, the air molecules next to it are reluctant to move and instead pile up on the face of the piston (Newton's first law). These compressed molecules now act as a spring, and release the pressure by jumping forward, creating a wave of compressed air molecules that moves through the tube. The potential energy has been converted to kinetic energy. These pressure waves move down the tube at a velocity of 344 m/s (at 20°C). If the piston oscillates at a *frequency* of, say, 10 cycles/s, there would be a series of pressure waves in the tube each 34.4 m apart. This relationship is expressed as

$$\lambda = \frac{c}{\nu}, \quad (22.1)$$

where

λ = wavelength (m),

c = velocity of the sound in a given medium (m/s), and

ν = frequency (cycles/s).

Sound travels at different speeds in different materials, depending on such physical properties of the material as its modulus of elasticity.

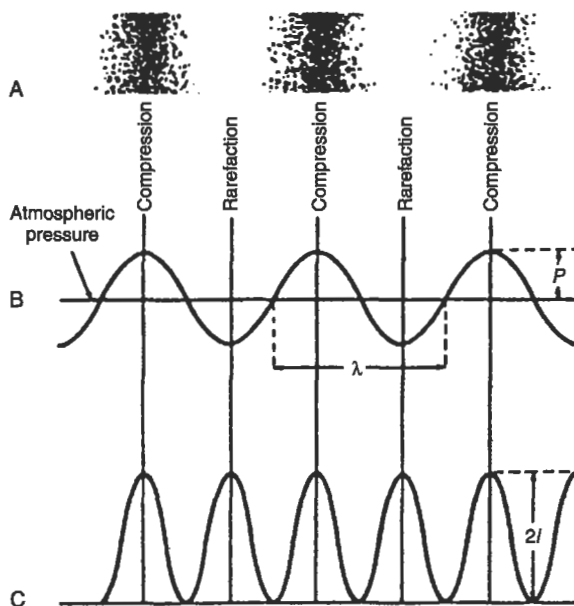


Figure 22-2. Sound waves. All parts of the figure show the spatial variation along the wave at a particular instant of time. (A) Regions of compression and rarefaction in the air. (B) Pressure wave; P = pressure amplitude ($P_{\text{rms}} = P/\sqrt{2}$) and λ = wavelength. (C) Intensity wave; the average intensity is denoted I .

EXAMPLE 22.1. In cast iron, sound waves travel at about 3440 m/s. What would be the wavelength of a sound from a train if it rumbles at 50 cycles/s and one listens to it placing an ear on the track?

$$\lambda = \frac{c}{\nu} = \frac{3440}{50} = 69 \text{ m.}$$

In acoustics, the frequency as cycles per second is denoted by the name hertz,¹ and written Hz. The range of frequency audible to the human ear is between 20 and 20,000 Hz. The middle A on the piano (concert pitch) is 440 Hz. The frequency is one of the two basic parameters that describe sound. *Amplitude*, how loud the sound is, is the other.

If the amplitude of a pressure wave of a sound with only one frequency is plotted against time, the wave is seen to produce a sinusoidal trace, as shown in Fig. 22-2.

¹In honor of the German physicist Heinrich Hertz (1857–1894).

All other sounds are made up of a number of suitable sinusoidal waves, as demonstrated originally by Fourier. Although both nonrandom and random combinations of sinusoidal waves can be pleasing to the ear, noise is usually a random combination.

Although the human ear is a remarkable instrument, able to detect sound pressures over seven orders of magnitude, it is not a perfect receptor of acoustic energy. In the measurement and control of noise, it is therefore important to know not only what a sound pressure is, but also to have some notion of how loud a sound *seems* to be. Before we address that topic, however, we must review some basics of sound.

SOUND PRESSURE LEVEL, FREQUENCY, AND PROPAGATION

Figure 22-2 represents a wave of pure sound: a single frequency. A sound wave is a compression wave, and the amplitude is a pressure amplitude, measured in pressure units like N/m^2 . As is the case with other wave phenomena, intensity is the square of the amplitude, or

$$I = P^2. \quad (22.2)$$

The intensity of a sound wave is measured in watts, a unit of power. When a person hears sounds of different intensities, the total intensity heard is not the sum of the intensities of the different sounds. Rather, the human ear tends to become overloaded or saturated with too much sound. Another statement of this phenomenon is that human hearing sums up sound intensities logarithmically rather than linearly. A unit called the *bel* was invented to measure sound intensity. Sound intensity level (IL) in bels is defined

$$\text{IL}_b = \log_{10} \left(\frac{I}{I_0} \right), \quad (22.3)$$

where

I = sound intensity in watts, and

I_0 = intensity of the least audible sound, usually given as $I_0 = 10^{-12} \text{ W}$.

The bel is an inconveniently large unit. The more convenient unit, which is now in common usage, is the *decibel* (dB). Sound intensity level in decibels is defined as

$$\text{IL} = 10 \log_{10} \left(\frac{I}{I_0} \right). \quad (22.4)$$

Since intensity is the square of pressure, an analogous equation may be written for sound pressure level (SPL) in decibels

$$\text{SPL (dB)} = 20 \log_{10} \left(\frac{P}{P_{\text{ref}}} \right), \quad (22.5)$$

where

SPL (dB) = sound pressure level (in dB),

P = pressure of sound wave, and

P_{ref} = some reference pressure, generally chosen as the threshold of hearing, 0.00002 N/m^2 .

These relationships are also derivable from a slightly different point of view. In 1825, E. H. Weber found that people can perceive differences in small weights, but if a person is already holding a substantial weight, that same increment is not detectable. The same idea is true with sound. For example, a 2 N/m^2 increase from an initial sound pressure of 2 N/m^2 is readily perceived, whereas the same 2 N/m^2 difference is not noticed if it is added to a background of 200 N/m^2 sound pressure. Mathematically, this may be expressed as

$$ds = K \frac{dW}{W}, \quad (22.6)$$

where

ds = is the minimum perceptible increase in a sensation (e.g., hearing),

W = the load (e.g., background SPL),

dW = the change in the load, and

K = a constant.

The integrated form of Eq. (22.6) is known as the Weber–Fechner Law

$$s = K \log_{10} W. \quad (22.7)$$

This idea is used in the definition of a *decibel*

$$\text{dB} = 10 \log_{10} \frac{W}{W_{\text{ref}}}, \quad (22.8)$$

where the power level W is divided by a constant W_{ref} , which is a reference value, both measured in watts.

Since the pressure waves in air are half positive pressure and half negative pressure, adding these would result in 0. Accordingly, sound pressures are measured as root-mean-square (rms) values, which are related to the energy in the wave. It may also be shown that the power associated with a sound wave is proportional to the square of the rms pressure. We can therefore write

$$\text{dB} = 10 \log_{10} \frac{W}{W_{\text{ref}}} = 10 \log_{10} \frac{P^2}{P_{\text{ref}}^2} = 20 \log_{10} \frac{P}{P_{\text{ref}}}, \quad (22.9)$$

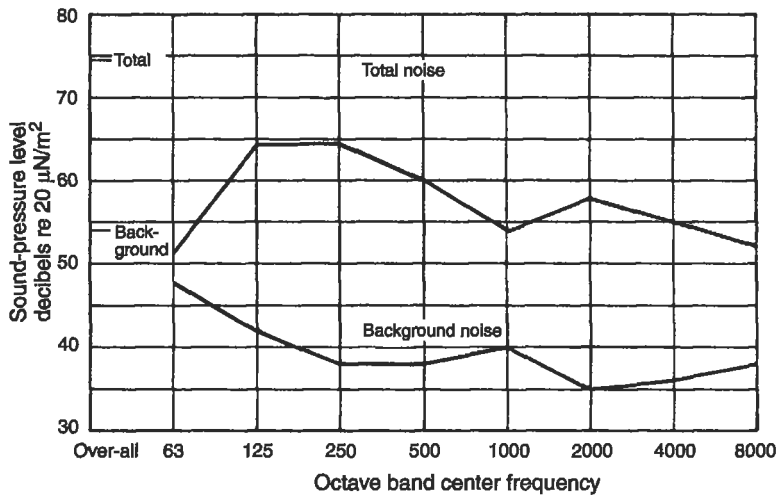


Figure 22-3. Typical analysis of a machine noise with background noise.

where P and P_{ref} are the pressure and reference pressure, respectively. If we define $P_{\text{ref}} = 0.00002 \text{ N/m}^2$, the threshold of hearing, we can define the SPL as

$$\text{SPL(dB)} = 20 \log_{10} \left(\frac{P}{P_{\text{ref}}} \right). \quad (22.5)$$

Although it is common to see the sound pressure measured over a full range of frequencies, it is sometimes necessary to describe a noise by the amount of sound pressure present at a specific range of frequencies. Such a *frequency analysis*, shown in Fig. 22-3, may be used for solving industrial problems or evaluating the danger of a certain sound to the human ear.

In addition to amplitude and frequency, sound has two more characteristics of importance. Both may be visualized by imagining the ripples created by dropping a pebble into a large, still pond. The ripples, analogous to sound pressure waves, are reflected outward from the source, and the magnitude of the ripples is dissipated as the ripples get farther away from the source. Similarly, sound levels decrease as the distance between the receptor and the source is increased.

In summary, then, the four important characteristics of sound waves are as follows:

- Sound pressure is the magnitude or amplitude of sound.
- The pitch is determined by the frequency of the pressure fluctuations.
- Sound waves propagate away from the source.
- Sound pressure decreases with increasing distance from the source.

The mathematics of adding decibels is a bit complicated. The procedure may be simplified a great deal by using the graph shown as Fig. 22-4. As a rule of thumb,

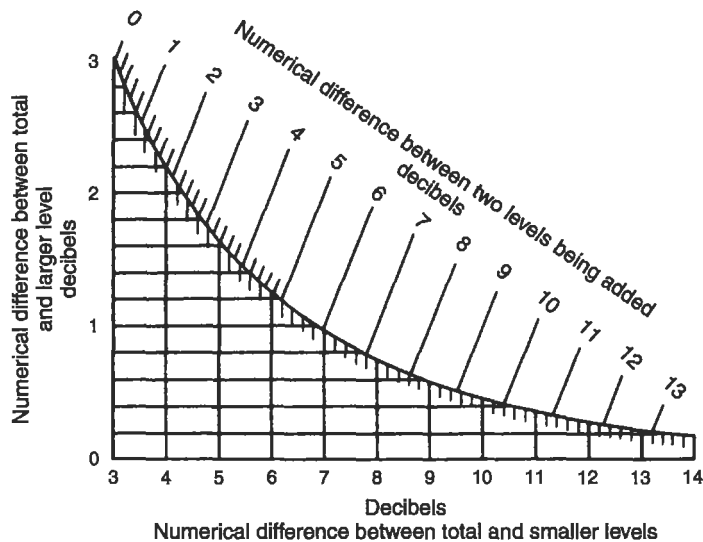


Figure 22-4. Chart for combining different sound pressure levels. For example: Combine 80 and 75 dB. The difference is 5 dB. The 5-dB line intersects the curved line at 1.2 dB; thus the total value is 81.2 dB. (Courtesy of General Radio.)

Table 22-1. Sound Pressure Levels for Speech Masking

Distance (ft)	Speech interference level (dB)	
	Normal	Shouting
3	60	78
6	54	72
12	48	66

adding two equal sounds increases the SPL by 3 dB and if one sound is more than 10 dB louder than a second sound, the contribution of the latter is negligible.

Background noise (or ambient noise) must also be subtracted from any measured noise. Using the above rule of thumb, if the SPL is more than 10 dB greater than the ambient level, the contribution may be ignored. The covering of a sound with a louder one is known as *masking*. Speech can be masked by industrial noise, for example, as shown in Table 22-1. These data show that an 80-dB SPL in a factory will effectively prevent conversation. Telephone conversations are similarly affected, with a 65-dB background making communication difficult and 80 dB making it impossible. In some cases, it has been found advisable to use *white noise*, a broad frequency hum from a fan, for example, to mask other more annoying noises.

EXAMPLE 22.2. A jet engine has a sound intensity level of 80 dB, as heard from a distance of 50 ft. A ground crew member is standing 50 ft from a four-engine jet. What SPL reaches her ear when the first engine is turned on? the second, so that two engines are running? the third? then all four?

When the first engine is turned on, the SPL is 80 dB, provided there is no other comparable noise in the vicinity. To determine, from the chart of Fig. 22-4, what the SPL is when the second engine is turned on, we note that the difference between the two engine intensity levels is

$$80 - 80 = 0.$$

From the chart, a numerical difference of 0 between the two levels being added gives a difference of 3 between the total and the larger of the two. The total SPL is thus

$$80 + 3 = 83 \text{ dB.}$$

When the third engine is turned on, the difference between the two levels is

$$83 - 80 = 3 \text{ dB,}$$

yielding a difference from the total of 1.8, for a total IL of

$$83 + 1.8 = 84.8 \text{ dB.}$$

When all four engines are turned on, the difference between the sounds is

$$84.8 - 80 = 4.8 \text{ dB,}$$

yielding a difference from the total of 1.2, for a total IL of 86 dB.

These characteristics ignore the human ear. We know that the ear is an amazingly sensitive receptor, but is it equally sensitive at all frequencies? Can we hear low and high sounds equally well? The answers to these questions lead us to the concept of *sound level*.

SOUND LEVEL

Suppose you are put into a very quiet room and subjected to a pure tone at 1000 Hz at a 40-dB SPL. If, in turn, this sound is turned off and a pure sound at 100 Hz is piped in and adjusted in loudness until you judge it to be “equally loud” to the 40-dB, 1000-Hz tone you had just heard a moment ago, you would, surprisingly enough, judge the 100-Hz tone to be equally loud when it is at about 55 dB. In other words, much more energy must be generated at the lower frequency in order to hear a tone at about the same perceived loudness, indicating that the human ear is rather inefficient for low-frequency tones.

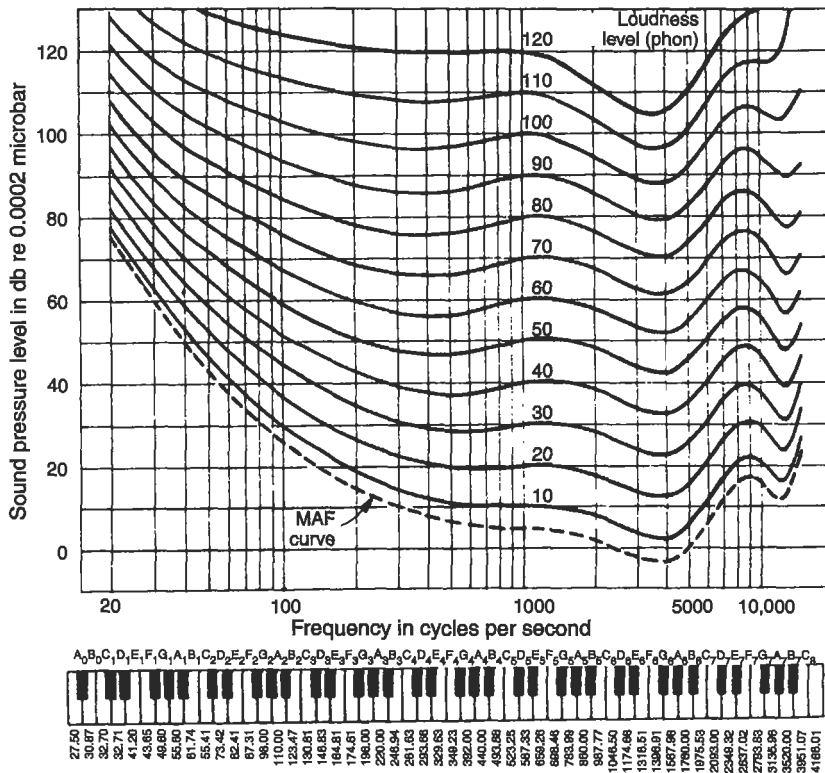


Figure 22-5. Equal loudness contours. (Courtesy of General Radio.)

It is possible to conduct such experiments for many sounds and with many people and to draw average equal loudness contours (Fig. 22-5). These contours are in terms of *phons*, which correspond to the sound pressure level in decibels of the 1000-Hz reference tone. Using Fig. 22-5, a person subjected to a 65-dB SPL tone at 50 Hz would judge this to be equally as loud as a 40-dB, 1000-Hz tone. Hence, the 50-Hz, 65-dB sound has a loudness level of 40 phons. Such measurements are commonly called *sound levels* and are not based on physical phenomena only, but have an adjustment factor, a "fudge factor," that corresponds to the inefficiency of the human ear.

Sound level (SL) is measured with a sound level meter consisting of a microphone, amplifier, a frequency-weighting circuit (filters), and an output scale, shown in Fig. 22-6. The weighing network filters out specific frequencies to make the response more characteristic of human hearing. Through use, three scales have become internationally standardized (Fig. 22-7). Figure 22-8 shows a typical hand-held sound level meter.

Note that the A scale in Fig. 22-7 corresponds closely to an inverted 40-phon contour in Fig. 22-5. Similarly, the response reflected by the B scale approximates an inverted 70-phon contour. The C scale shows an essentially flat response, giving equal

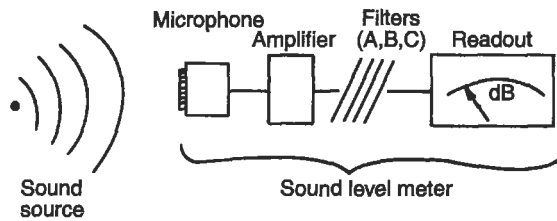


Figure 22-6. Schematic representation of a sound level meter.

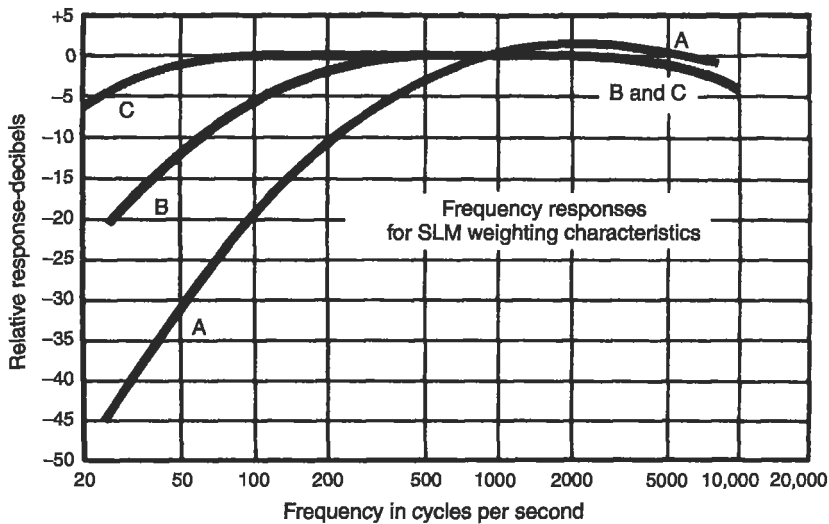


Figure 22-7. The A, B, and C filtering curves for a sound level meter.

weight to all frequencies, and approximates the response of the ear to intense sound pressure levels.

The results of noise measurement with the standard sound level meter are expressed in terms of decibels but with the scale designated. If on the A scale the meter reads 45 dB, the measurement is reported as 45 dB(A).

Most noise ordinances and regulations are in terms of dB(A), because this scale is a good approximation of human response for sounds that are not very loud. For very loud noise the C scale is a better approximation. However, because the use of multiple scales complicates matters, scales other than A are seldom used. In addition to the A, B, and C scales, a new D scale has been introduced to approximate human response to aircraft noise exclusively.

This brings up another complication involved in the measurement of noise. We do not generally respond in the same way to different types of noise, even though they may be equal in sound level. For example, a symphony orchestra playing middle C at 120 dB(A) and a jet engine at an equal 120 dB(A) will draw very different reactions



Figure 22-8. A typical sound level meter. (Courtesy of General Radio.)

from people. A plethora of parameters have been devised to respond to this discrepancy in reactions, all supposedly “the best” means of quantitatively measuring human response to noise. Among these are:

- Traffic noise index (TNI),
- Sones,
- Perceived noise level (PNdB),
- Noise and number index (NNI),
- Effective perceived noise level (EPNdB), and
- Speech interference level (SIL).

Measuring methods have proliferated because the physical phenomenon, a pressure wave, is related to a physiological human response (hearing) and then to a psychological response (pleasure or irritation). Along the way, the science becomes progressively more subjective.

Measuring some noises, particularly those commonly called *community noise*, such as traffic and loud parties, is complicated further. Although we have thus far treated noise as a constant in intensity and frequency with time, this is obviously not true for *transient noises* such as trucks moving past a sound level meter or a loud party near the measurement location. An unusual noise that may occur during a period of general quiet is an *intermittent noise* and has a still different effect on people.

MEASURING TRANSIENT NOISE

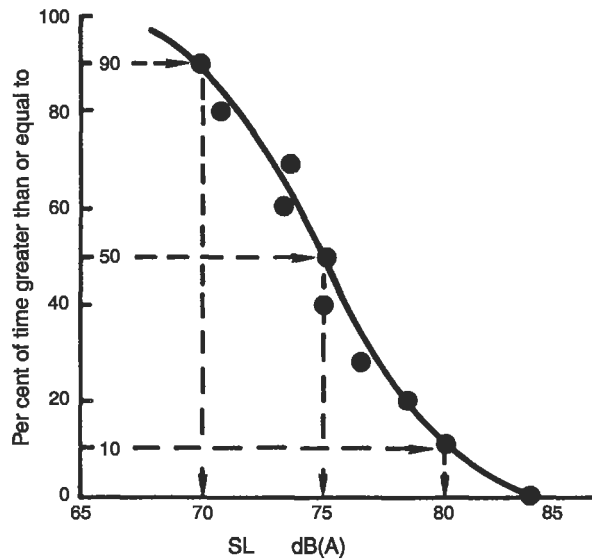
Transient noise is still measured with a sound level meter, but the results must be reported in statistical terms. The common parameter is the percent of time a sound level is exceeded, denoted by the letter L with a subscript. For example, $L_{10} = 70 \text{ dB(A)}$ means that 10% of the time the noise is louder than 70 dB as measured on the A scale. Transient noise data are gathered by reading the SL at regular intervals. These numbers are then ranked and plotted, and the L values are read off the graph.

EXAMPLE 22.3. Suppose the traffic noise data in Table 22-2 are gathered at 10-s intervals. These numbers are then ranked as indicated in the table and plotted as in Fig. 22-5. Note that since 10 readings are taken, the lowest reading (Rank no. 1) corresponds to a SL that is equaled or exceeded 90% of the time. Hence, 70 dB(A) is plotted versus 90% in Fig. 22-9.² Similarly, 71 dB(A) is exceeded 80% of the time.

²One might argue that since 70 dB(A) is the lowest value, it is the SL exceeded 100% of the time and one should plot 70 dB(A) vs 100%. The second value, 71 dB(A), is exceeded 90% of the time, etc. Either method is correct, and the error diminishes as the data increase.

Table 22-2. Sample Traffic Noise Data and Calculations for Example 22.3

Time (s)	dB(A)	Rank no.	% of time equal to or exceeded	dB(A)
10	71	1	90	70
20	75	2	80	71
30	70	3	70	74
40	78	4	60	74
50	80	5	50	75
60	84	6	40	75
70	76	7	30	76
80	74	8	20	78
90	75	9	10	80
100	74	10	0	84

**Figure 22-9.** Results from a survey of transient noise. The data (from Table 22-2) are plotted as percent of time the SL is exceeded vs the SL, in dB(A).

One widely used parameter for gauging the perceived level of noise from transient sources is the noise pollution level (NPL), which takes into account the irritation caused by impulse noises. The NPL is defined as

$$\text{NPL(dB(A))} = L_{50} + (L_{10} - L_{90}) + \frac{(L_{10} - L_{90})^2}{60}. \quad (22.10)$$

As defined earlier, the symbol L refers to the percentage of time the noise is equal to or greater than some value. The percentage is indicated by the subscript. L_{10} is thus the dB(A) level exceeded 10% of the time.

With reference to Fig. 22-9, L_{10} , L_{50} , and L_{90} are thus 80, 75, and 70 dB(A), respectively. These may then be substituted into the equation, and the NPL may be calculated. It is always advisable to take as much data as possible. The 10 readings in Table 22-2 are seldom sufficient for a thorough analysis. The percentages must be calculated on the basis of the total number of readings taken: if 20 readings are recorded, the lowest (Rank no. 1) corresponds to 95%, the second lowest, to 90%, etc.

THE ACOUSTIC ENVIRONMENT

The types of sound around us vary from a Beethoven concerto to the roar of a jet plane. Noise, the subject of this chapter, is generally considered to be unwanted sound, or the sound incidental to our civilization that we would just rather not have to endure. The intensities of some typical environmental noises are shown in Fig. 22-10. Laws against

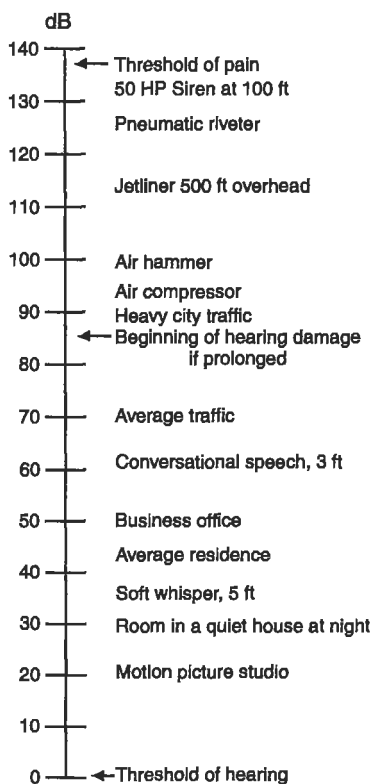


Figure 22-10. Environmental noise.

Table 22-3. OSHA Maximum Permissible Industrial Noise Levels

Sound level dB(A)	Maximum duration during any working day(h)
90	8
92	6
95	4
100	2
105	1
110	0.5
115	0.25

noise abound. Most local jurisdictions have ordinances against “loud and unnecessary noise.” The problem is that most of these ordinances are difficult to enforce.

In the industrial environment, noise is regulated by the federal Occupational Safety and Health Act (OSHA), which sets limits for noise in the workplace. Table 22-3 lists these limits. There is some disagreement as to the level of noise that should be allowed for an 8-h working day. Some researchers and health agencies insist that 85 dB(A) should be the limit. This is not, as might seem on the surface, a minor quibble, since the jump from 85 dB to 90 dB is actually an increase of about four times the sound pressure, remembering the logarithmic nature of the decibel scale.

HEALTH EFFECTS OF NOISE

The human ear is an incredible instrument. Imagine having to design and construct a scale for weighing just as accurately a flea or an elephant. This is the range of performance to which we are accustomed from our ears. Yet only recently have we become aware of the devastating psychological effects of noise. The effect of excessive noise on our ability to hear, on the other hand, has been known for a long time (Suter 1992). The human auditory system is shown in Fig. 22-11.

Sound pressure waves caused by vibrations set the eardrum (*tympanic membrane*) in motion. This activates the three bones in the middle ear. The *hammer*, *anvil*, and *stirrup* physically amplify the motion received from the eardrum and transmit it to the inner ear. This fluid-filled cavity contains the *cochlea*, a snail-like structure in that the physical motion is transmitted to tiny hair cells. These hair cells deflect, much like seaweed swaying in the current, and certain cells are responsive only to certain frequencies. The mechanical motion of these hair cells is transformed to bioelectrical signals and transmitted to the brain by the auditory nerves. Acute damage may occur to the eardrum, but this occurs only with very loud sudden noises. More serious is the chronic damage to the tiny hair cells in the inner ear. Prolonged exposure to noise of a certain frequency pattern may cause either temporary hearing loss that disappears in a few hours or days or permanent loss. The former is called *temporary threshold shift*,

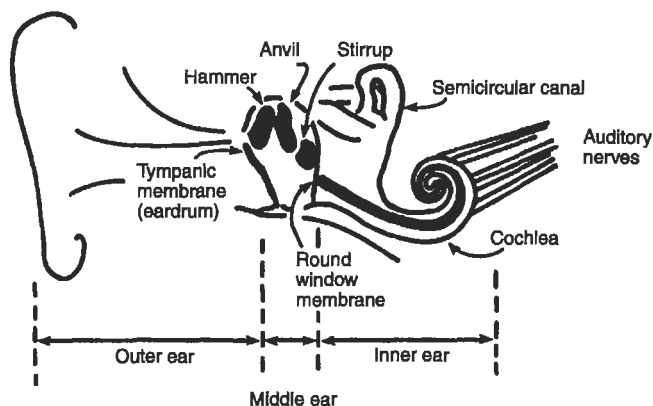


Figure 22-11. Cut-away drawing of the human ear.

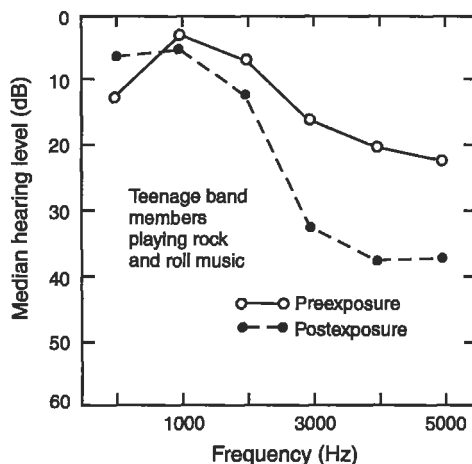


Figure 22-12. Temporary threshold shift for rock band performers. (Source: Data by the U.S. Public Health Service. See Still (1990).)

and the latter is known as *permanent threshold shift*. Literally, your threshold of hearing changes, so you are not able to hear some sounds.

Temporary threshold shift is generally not damaging to your ear unless exposure to the sound is prolonged. People who work in noisy environments commonly find that they hear worse at the end of the day. Performers in rock bands are subjected to very loud noises (substantially above the allowable OSHA levels) and commonly are victims of temporary threshold shift. In one study, the results of which are shown in Fig. 22-12, the players suffered as much as 15 dB temporary threshold shift after a concert.

Repeated noise over a long time leads to permanent threshold shift. This is especially true in industrial applications in which people are subjected to noises of a certain

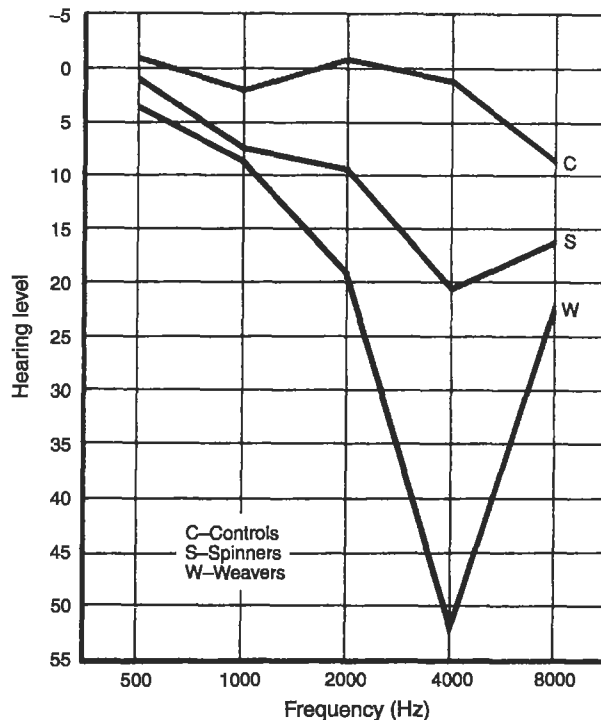


Figure 22-13. Permanent threshold shift for textile workers. (Source: W. Burns *et al.* "An Exploratory Study of Hearing and Noise Exposure in Textile Workers," *Ann. Occup. Hyg.* 7, 322 (1958).)

frequency. Figure 22-13 shows data from a study performed on workers at a textile mill. Note that the people who worked in the spinning and weaving parts of the mill, where noise levels are highest, suffered the most severe loss in hearing, especially at around 4000 Hz, the frequency of noise emitted by the machines.

As people get older, hearing becomes less acute simply as one of the effects of aging. This loss of hearing, called *presbycusis*, is illustrated in Fig. 22-14. Note that the greatest loss occurs at the higher frequencies. Speech frequency is about 1000 to 2000 Hz, where hearing loss is noticeable.

In addition to *presbycusis*, however, there is a serious loss of hearing owing to environmental noise. In one study (Taylor 1970), 11% of ninth graders, 13% of twelfth graders, and 35% of college freshmen had a greater than 15 dB loss of hearing at 2000 Hz. The study concluded that this severe loss resulted from exposure to loud noises such as motorcycles and rock music and that as a result "the hearing of many of these students had already deteriorated to a level of the average 65-year-old person."

Noise also affects other bodily functions including those of the cardiovascular system. Noise alters the rhythm of the heartbeat, makes the blood thicker, dilates blood vessels, and makes focusing the eyes difficult. It is no wonder that excessive

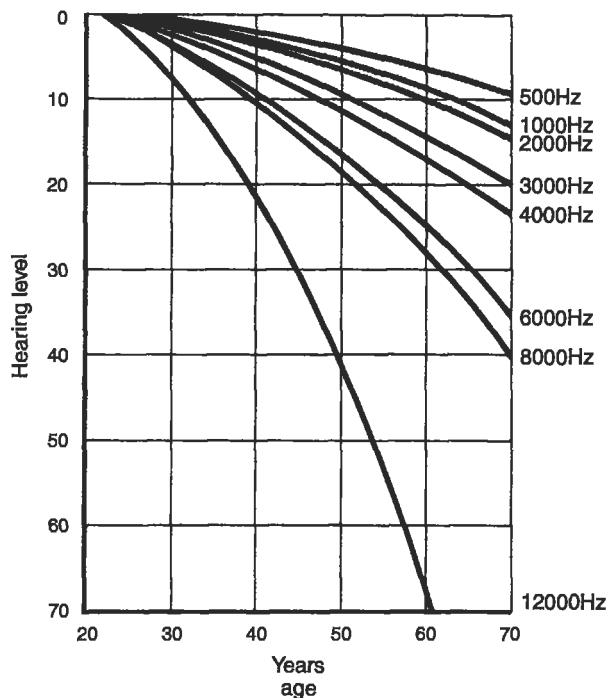


Figure 22-14. Hearing loss with age. (Source: R. Hinchcliffe, "The Pattern of the Threshold of Perception of Hearing and Other Special Senses as a Function of Age," *Gerontologica* 2, 311 (1958).)

noise has been blamed for headaches and irritability. All of these reactions are those that ancestral cave dwellers also experienced. Noise meant danger, and senses and nerves were "up," ready to repel the danger. In the modern noise-filled world, we are always "up," and it is unknown how much if any of our physical ills are due to our response to noise. We also know that we cannot adapt to noise in the sense that our body functions no longer react a certain way to excessive noise. People do not, therefore, "get used to" noise in the physiological sense.

In addition to the noise-we-can-hear problem, it is appropriate to mention the potential problems of very-high- or very-low-frequency sound, out of our usual 20 to 20,000 Hz hearing range. The health effects of these, if any, remain to be completely documented.

THE DOLLAR COST OF NOISE

Many case histories comparing patients in noisy and quiet hospitals point to increased convalescence time when the hospital was noisy, either from internal hospital activity or from external noise. This may be translated directly to a dollar figure.

Recent court cases have been won by workers seeking damages for hearing loss suffered on the job. The Veterans Administration spends millions of dollars every year for care of patients with hearing disorders. Other costs, such as lost sleep or sleeping pills, lost time in industry, and apartment soundproofing, are difficult to quantify, but examples abound. The John F. Kennedy Cultural Center in Washington spent \$5 million for soundproofing, necessitated by the jets using the nearby National Airport.

NOISE CONTROL

The control of noise is possible at three different stages of its transmission:

1. Reducing the sound produced,
2. Interrupting the path of the sound, and
3. Protecting the recipient.

When we consider noise control in industry, in the community, or in the home, we should keep in mind that all problems have these three possible solutions.

Industrial Noise Control

Industrial noise control generally involves the replacement of noise-producing machinery or equipment with quieter alternatives. For example, the noise from an air fan may be reduced by increasing the number of blades or the pitch of the blades and decreasing the rotational speed, thus obtaining the same air flow. Industrial noise may also be decreased by interrupting the path of the noise; for example, a noisy motor may be covered with insulating material.

A method of noise control often used in industry is protection of the recipient by distributing hearing protection devices (HPDs). These HPDs must be selected to have enough noise attenuation to protect against the anticipated exposures to noise. Great care must be taken to avoid interference with the ability to hear human speech and warning signals in the workplace.

Community Noise Control

The three major sources of community noise are aircraft, highway traffic, and construction. Construction noise must be controlled by local ordinances (unless federal funds are involved). Control usually involves the muffling of air compressors, jackhammers, hand compactors, etc. Since mufflers cost money, contractors will not take it upon themselves to control noise, and outside pressures must be exerted.

Regulating aircraft noise in the United States is the responsibility of the Federal Aviation Administration, which has instituted a two-pronged attack on this problem. First, it has set limits on aircraft engine noise and will not allow aircraft exceeding these limits to use the airports, forcing manufacturers to design engines for quiet operation as well as for thrust. The second effort has been to divert flight paths away from populated

areas and, whenever necessary, to have pilots use less than maximum power when the takeoff carries them over a noise-sensitive area. Often this approach is not enough to prevent significant noise-induced damage or annoyance, and aircraft noise remains a real problem in urban areas.

Supersonic aircraft present a special problem. Not only are their engines noisy, but the sonic boom may produce considerable property damage. Damage from supersonic military flights over the United States has led to a ban on such flights by commercial supersonic aircraft.

The third major source of community noise is traffic. The car or truck creates noise in a number of ways. The exhaust system, tires, engine, gears, and transmission all contribute to a noise level, while the very act of moving through the atmosphere creates wind noise. Elevated highways and bridges resonate with the traffic motion and amplify traffic noise. The worst offender on the highways is the heavy truck, which generates noise in all of these ways. The total noise generated by vehicles may be correlated directly to the truck traffic volume. Figure 22-15 is a typical plot showing sound level as a function of traffic volume (measured in number of trucks per hour). Clearly, truck volume is of great importance. It is important to note that this graph is plotted as "sound level exceeded 10% of the time." Peak sound levels could be a great deal higher.

A number of alternatives are available for reducing highway noise. First, the source could be controlled by making quieter vehicles; second, highways could be routed away from populated areas; and third, noise could be baffled with walls or other types of barriers.

Vegetation, surprisingly, makes a very poor noise screen, unless the screen is 50 yards or more deep. The opposing lanes of the Baltimore–Washington Parkway are separated in many places by 100–200 yards of fairly dense vegetation, which provides an excellent noise and light screen. Newer highways rarely have the luxury of so much right-of-way. The most effective buffers have been to lower the highway, or to build physical wood or concrete barriers beside the road and thus screen the noise. All of these have limitations: noise will bounce off the walls and create little or no noise

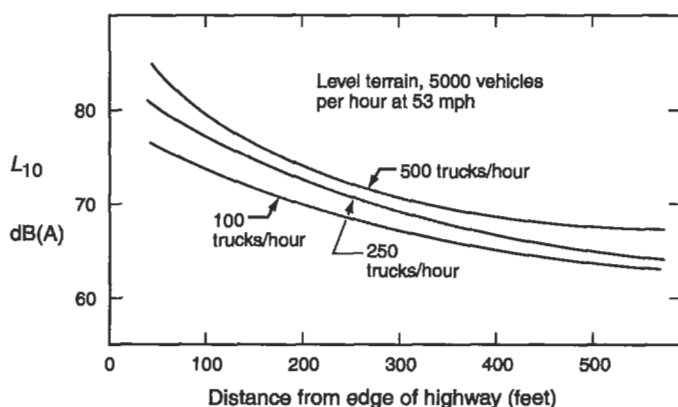


Figure 22-15. The effect of truck density and distance from a highway on noise.

Table 22-4. Design Noise Levels Set by the Federal Highway Administration

Land category	Design noise level, L_{10}	Description of land use
A	60 dB(A) exterior	Activities requiring special qualities of serenity and quiet, like amphitheaters
B	70 dB(A) exterior	Residences, motels, hospitals, schools, parks, libraries
	55 dB(A) interior	Residences, motels, hospitals, schools, parks, libraries
C	75 dB(A) exterior	Developed land not included in categories A and B
D	No limit	Undeveloped land

shadow, and walls hinder highway ventilation, thus contributing to buildup of CO and other pollutants from car and truck exhaust. The Department of Transportation has established design noise levels for various land uses, as shown in Table 22-4.

Noise in the Home

Private dwellings are getting noisier because of internally produced noise as well as external community noise. The list of gadgets in a modern American home reads like a list of New Year's Eve noisemakers. Some examples of domestic noise are listed in Table 22-5. Otherwise, similar products of different brands often will vary significantly in noise levels. When shopping for an appliance, it is just as important therefore to ask the clerk "How noisy is it?" as it is to ask "How much does it cost?"

The Eraring Power Station near Dove Creek, Australia, is an interesting example of the complexity of environmental pollution and control: all aspects of a problem must be considered before a solution is possible. During calm early mornings, residents complained of a loud "rumbling noise" coming from the power station 1.5 km away. The noise was measured in Dove Creek and for a single morning with no changes in the operation of the plant, the sound level increased from 43 dB at 5:30 AM to 62 dB at 6:00 AM. The problem: an atmospheric inversion "capturing" the noise and permitting it to reach the sleepy inhabitants of Dove Creek. The problem was solved when the duct work within the power station was attached to reactive dissipative silencers. The silencers, a series of chambers within the duct work system, were tuned to the predominant frequencies of the noise source. After a lengthy research project, the problem in Dove Creek was solved.

CONCLUSION

While noise was considered just another annoyance in a polluted world, not much attention was given to it. We now have enough data to show that noise is a definite

Table 22-5. Some Domestic Noisemakers

Item (distance from noise source)	Sound level, dB(A)
Vacuum cleaner (10 ft)	75
Quiet car at 50 mph (inside)	65
Sports car at 50 mph (inside)	80
Flushing toilet (5 ft)	85
Garbage disposal (3 ft)	80
Window air conditioner (10 ft)	55
Ringling alarm clock (2 ft)	80
Powered lawn mower (operator's position)	105
Snowmobile (driver's position)	120
Rock band (10 ft)	115

health hazard and should be numbered among our more serious pollutants. It is possible, using available technology, to lessen this form of pollution. However, the solution costs money, and private enterprise cannot afford to give noise a great deal of consideration until forced to by either the government or the public.

PROBLEMS

22.1 If an office has a noise level of 70 dB and a new machine emitting 68 dB is added to the din, what is the combined sound level?

22.2 Animals like dogs can hear sounds that have SPL *less than* 0. Show how this is possible.

22.3 Dogs can hear sounds at pressures close to $2 \times 10^{-6} \text{ N/m}^2$. What is this in decibels?

22.4 An air compressor emits pressure waves at 0.01 N/m^2 . What is the SPL in decibels?

22.5 Why was P_{ref} in Eq. (22.9) chosen at 0.00002 N/m^2 ? What if a mistake was made and P_{ref} should really have been 0.00004 N/m^2 ? This is a 100% error. How could this affect the numbers in Table 22-1?

22.6 Given the following data, calculate the L_{10} and the L_{50} .

Time (s)	dB(A)	Time (s)	dB(A)
10	70	60	65
20	50	70	60
30	65	80	55
40	60	90	70
50	55	100	50

22.7 Suppose your dormitory is 200 yards from a highway. What truck traffic volume would be "allowable" in order to stay within the Federal Highway Administration guidelines?